

A Review of *Explanation in Causal Inference: Methods for Mediation and Interaction*

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T. J. VanderWeele. *Explanation in Causal Inference: Methods for Mediation and Interaction*. New York, NY: Oxford University Press, 2015. 706 pp., \$84.80. ISBN 978-0-19-932587-0.

Causal inference as a subfield of statistics has increasingly gained popularity among applied researchers. This is unsurprising given that any attempt at understanding a social, psychological, or biological phenomenon involves some bold theories about causality. Scientific theories with sophistication often delineate the intermediate process as well as the external or internal conditions of an important phenomenon. “Mediation” and “moderation” have been buzzwords in many disciplines. Empirical applications designed for theory testing, however, tend to conclude with ambiguous terms. This is especially true in the absence of experimental manipulations of the intermediate process or of the presumed conditions.

So what does causal inference have to offer to applied research? Quite a lot.

Firstly, it offers a potential outcomes framework for defining every causal effect of interest without invoking arbitrary model-based assumptions. In other words, a causal effect can be clearly defined *without* a regression model.

Secondly, it makes explicit all the assumptions required for identifying the causal effect of interest. Rather than taking the assumptions for granted, it forces applied researchers to seriously scrutinize these assumptions in every study context and assess the potential consequences of violations.

And thirdly, it gives rise to a proliferation of research designs and analytic methods in competition for identifying and estimating the causal effects with fewer or weaker assumptions.

In this context, Tyler VanderWeele’s (2015) book *Explanation in Causal Inference: Methods for Mediation and Interaction* arrives timely among other titles. After contributing numerous journal articles on related topics, mostly in the field of epidemiology, the author must have naturally found it imperative to organize the materials into a coherent and comprehensive frame. At least for me, instead of spending hours searching for the articles online, having the 706-page book handy is a lifesaver.

After an introduction, the book devotes seven chapters to mediation analysis, five to interaction analysis, and three to synthesis and spillover effects. The

allocation of weight to the three major topics reflects the configuration of the author's ambitious research agenda, while the rigor in his reasoning is in full display in the nearly 200 pages of appendices supplying technical details and proofs. The first part of the book focuses on mediation analysis addressing the question of "why." A mediation analysis investigates the intermediate process of an observed phenomenon, which is at the core of any scientific explanation. Two leading chapters, one introducing the basic concepts of and the regression-based approaches to mediation and the other describing sensitivity analysis for mediation, prepare readers for a range of mediation-related topics. These include survival data, multiple mediators, and longitudinal data with time-varying exposures and mediators. The second part of the book presents methods for analyzing "interaction." Such an analysis examines whether various conditions moderate the relationship between a cause and an effect and whether multiple exposures jointly cause a phenomenon. The last part of the book attempts to synthesize the methods for mediation analysis and those for interaction analysis. The author also allocates a chapter to a discussion of "interference," which is also called "spillover effects" that often arise due to social interactions.

Upon reading the book, I believe it cannot be a pure coincidence that its topics mirror those of my own book *Causality in a Social World: Moderation, Mediation, and Spill-over* (Hong, 2015). Tyler and I exchanged research ideas from time to time and collaborated in a study at one point (VanderWeele, Hong, Jones, & Brown, 2013). Interestingly, we were unaware of each other's book writing plans. What I can conclude is that these two books are related, complementary, but irredundant. Here is what I have found to be unique in Tyler's book.

The book explicitly builds on readers' prior training in multiple regression and focuses on a regression-based approach to mediation and moderation analyses. This emphasis provides coherence to the entire text. In the section that introduces causal mediation analysis, readers who are familiar with path analysis and linear structural equation modeling (SEM) would feel reassured that, after paying enough attention to the identification assumptions, in many cases, this well-known approach may actually generate causally meaningful results. The key assumptions, in their essence, require that there is no omitted confounding after one has adjusted for a set of observed covariates. Another nontrivial assumption is that the treatment and the mediator are additive in predicting the outcome. The book stresses that, within the regression framework, one may accommodate a Treatment $\times$ Mediator interaction. As a result, the indirect effect and the direct effect that decompose the total effect will each be a function of multiple regression coefficients. These functions take a form similar to that of "Oaxaca decomposition" (Oaxaca, 1973) known to some social scientists. This section further shows some limited capacity that the regression framework holds for accommodating a nonlinear mediator model or a nonlinear outcome model.

To my knowledge, Holland (1988) was the first to place path analysis and SEM under the potential outcomes framework and explicate the entire set of

assumptions implied by these models. Jo (2008) and Sobel (2008) made further attempts to clarify the challenges to causal mediation analysis and evaluated possible solutions when SEM is employed. These earlier scholarly contributions, along with the current book, should together provide a strong antidote to misuse of SEM often prevalent in applied research.

Yet I feel that an important discussion is lacking here. When multiple regression is utilized in causal analysis to adjust for confounding, the data analyst is often unaware of the consequences of misspecifying the treatment model (when the treatment is not randomized), the mediator model, or the outcome model. For example, if a covariate confounds the mediator–outcome relationship differently across the treatment conditions, omitting the Treatment $\times$ Covariate interaction in the mediator model will bias the analytic results. In my view, this is a major potential pitfall of regression that can hardly be overemphasized. A simple statement that each model needs to be correctly specified may not serve as enough a precaution against malpractice (cf. Cochran, 1965; Hong, 2010a). I tend to think that readers would be helped by seeing various forms of potential incorrect specifications and the consequent bias. It would be at least worthy of noting that omitted nonlinear and nonadditive terms of covariates constitute a subset of omitted confounders.

Because a causal mediation or moderation analysis is valid only when the corresponding identification assumptions hold, and because the assumptions of no omitted confounders rarely hold, assessing the consequence of violating an identification assumption is a necessary step in causal inference. Measurement errors in predictors of a regression model bring another source of bias. The book is especially thorough in emphasizing sensitivity analysis for each mediation or moderation analysis. Given that sensitivity analysis has not been a standard practice in a majority of applied research, placing a great amount of emphasis on this topic may generate a much needed intervention. Yet there is one caveat: The approach to sensitivity analysis presented in the book is again often based on the regression framework. Model-based assumptions are invoked for reducing the number of sensitivity parameters and for simplifying the implementation. Ironically, this seems to leave the analyst in a tail-chasing game—that is, one must heavily rely on one set of assumptions in an attempt to quantify the bias due to the violation of another set of assumptions.

In addition to the above highlights, the book's broad coverage of content reflects the author's wide-ranging research interests and solid grasp of the substantive and methodological developments in epidemiology and beyond. For example, in the discussion of mediation analysis, the book additionally introduces to readers the simulation-based approach proposed by Imai and colleagues (Imai, Keele, & Tingly, 2010; Imai, Keele, & Yamamoto, 2010) and the semi-parametric weighting approach that I initially developed (Hong, 2010b) and others have embraced (see Lange, Vansteelandt, & Bekaert, 2012). Examinations of sufficient causes and synergism help shift the attention from "effects" to

“causes,” which promise to enrich the conceptual discussions of causality that have previously placed all the bets on the former. And finally, theorizing the spillover process in mass vaccination, in social organizations, and in formal and informal networks inserts new challenges and excitements to this literature.

According to the author, the book grew out of a course for second-year graduate students in applied disciplines including epidemiology and the social and behavioral sciences. Hence, readers of a similar background should find the materials accessible and relevant to their interests. This should be true especially in Chapters 2, 3, and 9 that lay out the foundational concepts and methods for analyzing mediation and interaction and even provide SAS, SPSS, and Stata macros for implementation. The book also aims to direct the attention of statisticians and methodologists to important areas for further inquiry. The technical appendix primarily serves the interest of this latter group.

The author has been a major contributor to the advances of causal inference methods for mediation and moderation. In each of his published articles that I have read, he always patiently walks the reader through the conceptual and mathematical definition of every causal effect in terms of counterfactual outcomes. He then shows how the counterfactual outcomes can be linked to observable data under a series of identification assumptions and how the identification can be implemented, often through a regression approach, by invoking an additional set of model-based assumptions. To my surprise, in order to make this book broadly accessible, he deliberately decided to avoid using the notation representing counterfactual outcomes and even advise readers to skip sections in which such a notation is employed. As a result, the counterfactual nature of a causal effect is communicated in words only, and these words are typically embedded in a lengthy text. In contrast, equations that represent these causal effects each as a function of regression coefficients are displayed with glaring salience. This seems to be a risky decision. I am afraid that readers without training in causal inference would mistakenly regard these regression functions as formal definitions of the causal effects. On the contrary, a causal effect must be defined *without* a regression model and hence *without* any model-based assumptions! I would advise that such readers task themselves with locating the definition of each causal effect in the text, highlighting the words with a marker, and carefully digesting the meaning of every counterfactual outcome.

Many people have already put a copy of this book on their shelves or desks. That itself clearly speaks to the demand for rigorous causal approaches to everyday practice in applied research. I applaud Tyler for being a pioneer in promoting this fundamental transformation in biological, behavioral, and social sciences. I would be eager to see the next edition of this book.

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